

FIR High Pass FilterS using Windowing Techniques 5.1. Design and Analysis of an FIR High-Pass Filter Using the Rectangular Window Technique in MATLAB.

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Filter Specifications
```

```
Fs = 1000; % Sampling Frequency (Hz)
```

```
Fc = 150; % Cutoff Frequency (Hz)
```

```
N = 50; % Filter Order
```

```
%% Normalize Cutoff Frequency
```

```
Wn = Fc/(Fs/2);
```

```
%% Design FIR High-Pass Filter using Rectangular Window
```

```
b = fir1(N, Wn, 'high', rectwin(N+1));
```

```
a = 1;
```

```
%% Display Filter Information
```

```
disp('FIR High-Pass Filter using Rectangular Window');
```

```
disp(['Filter Order = ', num2str(N)]);
```

```
disp('Filter Coefficients:');
```

```
disp(b);
```

```
%% Frequency Response
```

```
figure;
```

```
freqz(b,a,1024,Fs);
```

```
title('Frequency Response of FIR High-Pass Filter (Rectangular Window)');
```

```
%% Magnitude and Phase Response
```

```
[H,f] = freqz(b,a,1024,Fs);
```

```
figure;
```

```

subplot(2,1,1);
plot(f,20*log10(abs(H)),'b','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response');
subplot(2,1,2);
plot(f,unwrap(angle(H))*180/pi,'r','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response');
%% Impulse Response
figure;
impz(b,a);
grid on;
title('Impulse Response');
%% Filter Coefficients
figure;
stem(0:N,b,'filled');
grid on;
xlabel('Sample Number');
ylabel('Amplitude');
title('Filter Coefficients');
%% Pole-Zero Plot
figure;

```

```
zplane(b,a);

grid on;

title('Pole-Zero Plot');

%% Generate Test Signal

t = 0:1/Fs:1;

% Low Frequency Signal (50 Hz)

x1 = sin(2*pi*50*t);

% High Frequency Signal (250 Hz)

x2 = 0.5*sin(2*pi*250*t);

% Combined Signal

x = x1 + x2;

%% Apply High-Pass Filter

y = filter(b,a,x);

%% Time Domain Signals

figure;

subplot(2,1,1);

plot(t,x,'b');

grid on;

xlabel('Time (s)');

ylabel('Amplitude');

title('Input Signal');

subplot(2,1,2);

plot(t,y,'r');

grid on;

xlabel('Time (s)');

ylabel('Amplitude');
```

```

title('Filtered Output Signal');

%% FFT Analysis

L = length(x);
X = abs(fft(x));
Y = abs(fft(y));
faxis = (0:L-1)*(Fs/L);

figure;
subplot(2,1,1);
plot(faxis,X,'b');
xlim([0 500]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Input Signal Spectrum');
subplot(2,1,2);
plot(faxis,Y,'r');
xlim([0 500]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Filtered Signal Spectrum');

%% Display Parameters

fprintf('\nSampling Frequency : %.0f Hz\n', Fs);
fprintf('Cutoff Frequency : %.0f Hz\n', Fc);
fprintf('Filter Order : %d\n', N);
fprintf('Window Used : Rectangular Window\n');

```

OUTPUTS:





